

Luiss

Libera Università Internazionale
degli Studi Sociali Guido Carli

Homework Exercises on Binary Numbers and Boolean Logic

Introduction to Computer Programming

Management and Artificial Intelligence

LUISS



Binary Number Exercises

Decimal 387 to Binary

Convert the decimal number **387** to binary:

Decimal 387 to Binary

Convert the decimal number **387** to binary:

Result: $(110000011)_2$

Binary 110110101011 to Decimal

Convert the binary number **110110101011** to decimal:

Decimal 2048 to Hexadecimal

Convert the decimal number **2048** to hexadecimal:

Hex BADCODE to Binary

Convert the hexadecimal number **BADCODE** to binary:

Text TOTTI to Binary

Convert the text "TOTTI" to binary using the ASCII table

Binary to text

Convert 0100110001010101010010010101001101010011 into text using ASCII:

Boolean Logic Exercises

Formula $(A \wedge \neg B \wedge C) \vee (\neg A \wedge B \wedge \neg C) \vee (A \wedge B \wedge C)$

Formula $(\neg A \wedge \neg B \wedge \neg C) \vee (A \wedge B \wedge C) \vee (A \wedge \neg B \wedge \neg C)$

Formula $(A \wedge B) \vee (\neg A \wedge \neg B \wedge C) \vee (A \wedge \neg B \wedge \neg C)$

Formula $(A \wedge B \wedge C) \vee (\neg A \wedge \neg B \wedge \neg C)$ is equivalent to what?

Formula $(A \vee B) \wedge \neg(A \wedge B)$ is a tautology?

Formula $\neg A \wedge \neg B \wedge A$ is a contradiction?

